

Quiz # 2 (OOP)

Name: _____

Time Allowed: 15 min

BS DS Fall 2025 Afternoon

Roll number: _____

Total Marks: 20

Question#1 (3+3+4=10 Marks)

CLO3

Write down the output of the following codes:

<pre>int main() { int arr[5]={3,6,8,2,4}; int *p *q; int a ,b; p= arr+1; q=arr+3; cout << *p << " " << *q << endl; a=*(p+3); b=(*q)+3; cout << a << " " << b << endl; p=p+2; q=q-*q; cout << *p << " " << *q << endl; return 0; }</pre>	<pre>6 2 4 5 2 6</pre>
<pre>int main(){ int x = 50, y = 60, z = 70; int * ptr; cout << x << " " << y << " " << z << endl; ptr = &x; *ptr *= 10; ptr = &y; *ptr += 5; ptr = &z; *ptr *= 2; cout << x << " " << y << " " << z << endl; return 0; }</pre>	<pre>50 60 70 500 65 140</pre>
<pre>void inc_count(int *count_ptr) { (*count_ptr)++; } int main() { int count = 0; while (count < 10) inc_count(&count); cout << count << endl; return 0; }</pre>	<pre>10</pre>

Question#2 (10 Marks)

CLO3

Considering the following code:

<pre>struct Book { char name[50]; char publisher[50]; int ISBNNumber; };</pre>	<pre>struct Library { Book * data; int noOfBooks; int capacity; };</pre>
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Write down the definition of the following function, it will Increase the capacity of the library by s (default value is 5). Ensure that the existing data is preserved correctly.

void resize(Library& ly , int s=5);

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void resize(Library& ly, int s = 5)
{
    // Calculate new capacity
    int newCapacity = ly.capacity + s;

    // Allocate new dynamic array with increased capacity
    Book* newData = new Book[newCapacity];

    // Copy existing books into new array
    for (int i = 0; i < ly.noOfBooks; i++)
    {
        newData[i] = ly.data[i];
    }

    // Free old memory
    delete[] ly.data;

    // Update pointer and capacity
    ly.data = newData;
    ly.capacity = newCapacity;
}
```